POOJA KIRAN | SECURITY ENGINEERING

Version-Aware Normalization of Findings Against MITRE ATT&CK v19

Typed Data Models and Revocation-Resolving Technique Lookup for Security Tooling

Source inventory: 591ad98911eebe3f8a681db2accc9a653d951320

Test snapshot: 22cc9ee3e08406a446bf500f84ea1d8fb53a6a6d

CI run 36783662812 | initiated 2026-09-30 UTC | MIT

https://github.com/poojakira/attack-v19-core

165
tests passed in cited CI

63.83%
statement coverage at cited CI

29
source remap entries, including 22 release revocations

46
new-technique entries in repository migration table

CONTRIBUTION

- A typed ATT&CK v19 data layer that normalizes findings across version churn by resolving revoked/renamed technique IDs through a maintained remap table over SHA-256-verified STIX data, with indexed lookup for downstream security tooling.

IMPLEMENTED METHOD

- 1. Download/validate ATT&CK STIX data with SHA-256 verification.
- 2. Parse techniques/tactics/platform metadata into typed structures.
- 3. Resolve deprecated/revoked IDs through the v19 remap table.
- 4. Build in-memory indexes for ID, tactic, platform, and keyword lookup.
- 5. Regression-test migration and data-loader behavior across supported Python versions.

LIMITATIONS AND EVIDENCE BOUNDARY

- ATT&CK mapping/normalization does not prove an attack occurred.
- O(1) dictionary/index lookup describes the in-memory data structure, not end-to-end system latency.
- Future ATT&CK releases require new migration/remap evidence.

SCOPED EVIDENCE

- Cited Python 3.12 CI: 165 passed; 63.83% statement coverage, same test job.
- Current source inventory: 22 release-revocation entries, 29 total remaps and 46 V19_NEW_TECHNIQUES entries. The earlier 48 figure was not the implemented table count.
- The alleged release-wide total of 48 was not independently verified from official STIX/release evidence; this poster reports the repository table only.
- Loader/index code performs pinned STIX validation and lookup; local full STIX-dependent execution remains blocked by missing data.

REPRODUCTION

- git checkout 22cc9ee3e08406a446bf500f84ea1d8fb53a6a6d
- Install the documented development dependencies; run python -m pytest tests/ -q. Use the cited job configuration when reproducing coverage.
- Benchmark data, adapters and environment requirements are separate; consult poster/13_reproducibility.md and the claim ledger.

Dated tests and committed fixtures do not establish universal detection, production deployment or an SLO.